CAM-IMX415-4K User Manual

Date	Version
2025/12/03	V1.0

1. Description

1.1 Overview

The CAM-IMX415-4K is a dedicated true 4K camera module developed exclusively for the Raspberry Pi platform (Raspberry Pi 5 / 4B / 3B+ / 3B / Zero 2W / Zero W / CM4 / CM5).

It utilizes the authentic Sony STARVIS IMX415 8.46-megapixel back-illuminated sensor and ships with raspberry pi os built in tuning file, paired with a high-quality lens offering a field of view (H 70.2° / V 39.5° / D 80.6°,F/1.8,Focal Length 4.9mm).

Fully validated mode is **3864 × 2192 (true 4K) @ 30 fps**,Ideal for 4K AI vision, high-resolution surveillance, industrial AOI, machine vision, scientific imaging and long-term 4K time-lapse projects requiring broad coverage with the specified FOV.

1.2 Key Features

- Genuine Sony STARVIS IMX415 sensor – real hardware 4K
- Officially supported mode: **3864 × 2192 @ 30 fps**
- Starlight-level full-color imaging down to 0.01 Lux
- Lens field of view: H 70.2° / V 39.5° / D 80.6°
- Built IR filter
- Native support through official libcamera / rpicam-apps (dtoverlay=imx415 + tuning file required)
- 2-lane CSI-2 bandwidth utilization on Raspberry Pi 5
- Low power consumption

2. Technical Specifications

2.1 Sensor

Item	Specification
Sensor Model	Sony IMX415 (STARVIS series)
Sensor Type	1/2.8" Back-illuminated stacked CMOS
Total Pixels	3864 (H) × 2192 (V) = 8.46 Megapixels
Raspberry Pi Usable Area	3864 × 2192
Pixel Size	1.45 µm × 1.45 µm
Shutter Type	Electronic rolling shutter
Output Format	RAW10 (default) / RAW12
Max SNR	37.1 dB
Dynamic Range	71.3 dB (standard), up to 110 dB (DOL-HDR)
Sensitivity	0.01 Lux @ F1.8 (full-color starlight)

2.2 Lens Optics

Follow Chapter 3.3

2.3 Electrical

Item	Specification
Interface	MIPI CSI-2 (2-lane on Pi 5, 2-lane on others)
Power Supply	3.3 V (directly from Raspberry Pi CSI port)

2.4 Mechanical

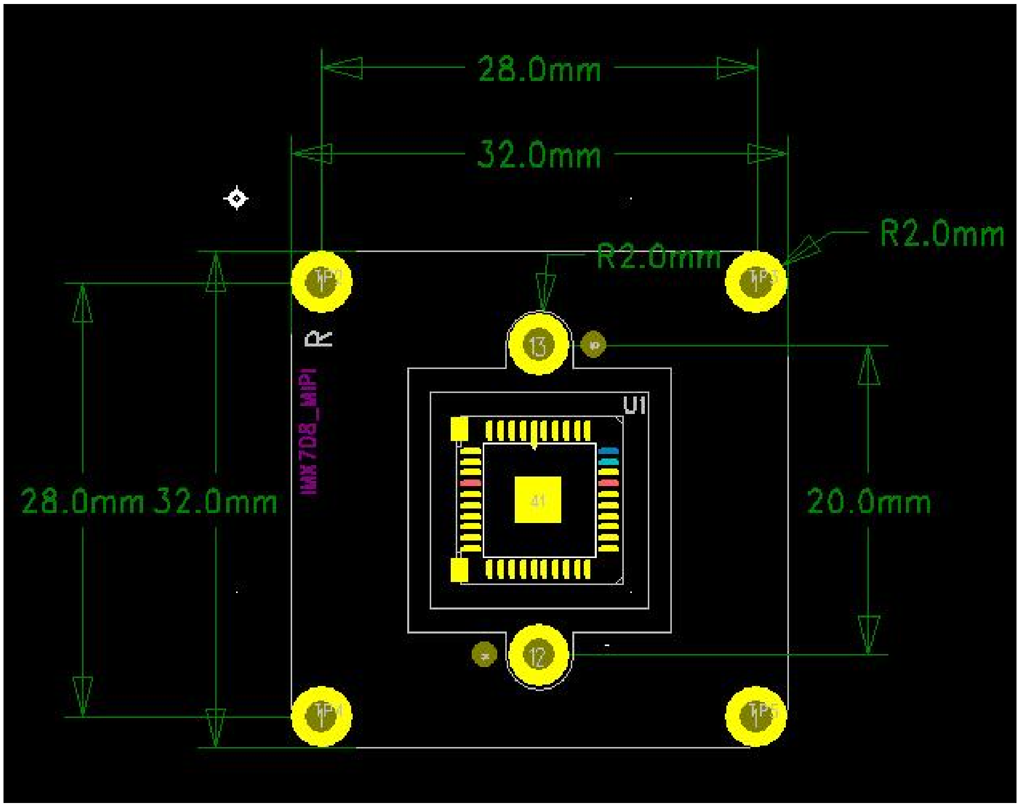
Item	Specification
Module Dimensions	32 × 32 mm
Weight	15g
Flex Cable	15PIN-15PIN 1.5mm 15PIN-22PIN 1mm
Mounting Holes	4 × M2

2.5 Environmental

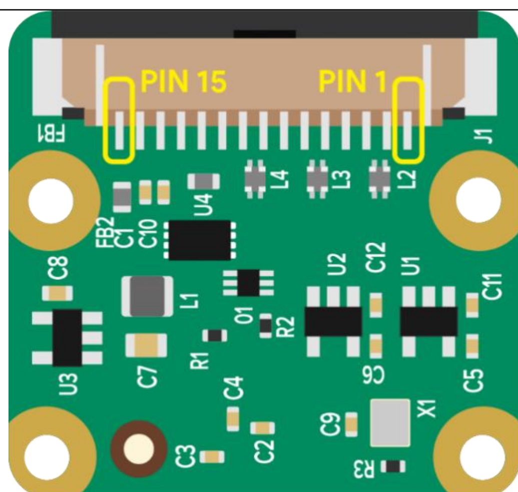
Item	Specification
Operating Temperature	–20 °C to +70 °C
Storage Temperature	–30 °C to +85 °C
Humidity	20 %–85 % RH (non-condensing)

3 Hardware

3.1 Camera Size



3.2 Camera Module Pins Out



Pin #	Name	Description

1	GND	Ground
2	CAM_D0_N	MIPI Data Lane 0 Negative
3	CAM_D0_P	MIPI Data Lane 0 Positive
4	GND	Ground
5	CAM_D1_N	MIPI Data Lane 1 Negative
6	CAM_D1_P	MIPI Data Lane 1 Positive
7	GND	Ground
8	CAM_CK_N	MIPI Clock Lane Negative
9	CAM_CK_P	MIPI Clock Lane Positive
10	GND	Ground
11	CAM_IO0	Power Enable
12	CAM_IO1	LED Indicator
13	CAM_SCL	I2C SCL
14	CAM_SDA	I2C SDA
15	CAM_3V3	3.3V Power Input

3.3 Lens Information

5 主要参数（Main Items）

5.1 光学参数规格（Optical Specifications）

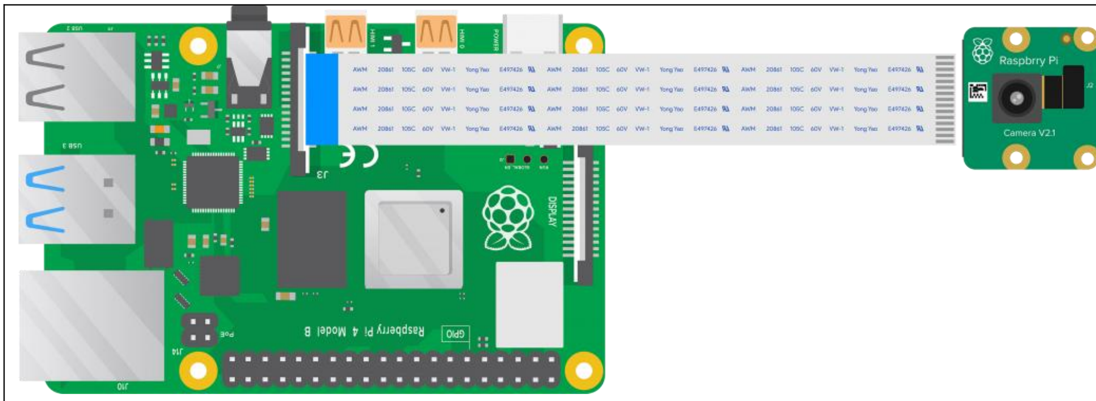
M12 4mm 1/1.8

NO.	项目(Items)		具体规格（Specification）		
1	F No.		1.65±10%		
2	焦距(Focal-Length)		4.9±5%		
3	光学后焦 (Optical Back Focal Length)		6.41±0.2 (in air)		
4	机械后焦 (Mechanical Back Focal Length)		6.17±0.2 (in air)		
5	镜头总长(TTL)		22.28±0.2 (in air)		
6	像面大小（Image circle)		Φ9.2（MAX)		
7	镜片构成（Lens structure)		4G4P		
8	接口 (Mount)		M12*P0.5		
9	镜头与底座螺纹配合扭力 (Clamp Force)		60-600gf.cm		
10	视场角(FOV)	sensor型号	H(水平)	V(垂直)	D(对角)
		1/1.8"(16:9)	86. 7°	48. 8°	99. 5°
		1/2.5"(16:9)	70. 2°	39. 5°	80. 6°
		1/2.7"(16:9)	65°	36. 6°	74. 6°
11	光学畸变（Optical Distortion)		1/1.8"	1/2.5"	1/2.7"
			-25.6%	-17.6%	-14.8%
12	相对亮度（Relative Illumination)		1/1.8"	1/2.5"	1/2.7"
			40%	58%	60%
13	最大主光线夹角（CRA)		1/1.8"	1/2.5"	1/2.7"
			16.2°	12.3°	11.9°
14	近摄距（M.O.D)		0.6m		
15	解像标准（Resolution)		分辨率（Resolution): 3840*2160 （8MP）		
16	重量 (Weight)		/		
17	操作方法 (Operation)		聚焦 (Focus)	手动（Manual)	
			光圈 (Iris)	固定（Fixed)	
18	环保&安全（HSF&Safety)		RoHS		

Apply for IMX415 Sensor Module

Sensor	Sensor size / typical mode	Focal Length	Aperture	Horizontal FOV	Vertical FOV	Diagonal FOV
IMX335	1/2.8" (≈ 1/1.8" mode, 16:9)	4.9 mm (±5%)	f/1.8	86.7°	48.8°	99.5°
IMX415	1/2.8" (≈ 1/2.5" mode, 16:9)	4.9 mm (±5%)	f/1.8	70.2°	39.5°	80.6°

3.4 Connection



4. Quick Start Guide

4.1 Edit config file

```
sudo nano /boot/firmware/config.txt   # Pi 5
# or
sudo nano /boot/config.txt            # Pi 4/3/Zero
```

4.2 Add these two lines at the bottom:

```
camera_auto_detect=0
dtoverlay=imx415,cam0 #PI5,CSI0
Or
dtoverlay=imx415,cam1 #PI5,CSI1
```

Save and reboot

```
sudo reboot
```

4.3 Test commands:

```
rpicam-hello -t 0                    # Live preview
```

```
rpicam-hello -t 0 --camera 0                    # Point channel if Dual camera

rpicam-hello --list-cameras                     # list cameras

rpicam-hello --list-cameras -v                  # list camera rw value


rpicam-still -o 4k.jpg --width 3864 --height 2192        # Capture 4K photo

rpicam-vid -t 20000 --width 3864 --height 2192 -o 4k_30fps.h264   # Record 20 s 4K video
```

5 Official Documentation

- Camera Description on Raspberry Pi
- Camera Software On Raspberry Pi